

Personalized Electrocardiographic and HRV Dynamics for Acute Stress Detection: A Leave-One-Subject-Out Benchmark and Bare-Metal Edge IoMT Implementation

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Abstract—Automated classification of acute psychological stress from non-invasive wearable electrocardiography (ECG) is a fundamental problem in physiological computing, affective state recognition, and wearable Internet of Medical Things (IoMT). A central barrier to cross-subject generalization is inter-individual baseline heterogeneity: resting heart rate and basal heart rate variability (HRV) metrics vary widely across individuals due to genetic, cardiorespiratory fitness, and circadian factors, causing uncalibrated global classifiers to degrade substantially on unseen subjects. Furthermore, resource constraints and exposure of physiological telemetry create additional challenges for edge-node wearable deployment. In this work, we present an end-to-end reproducible research pipeline benchmarked across all 15 subjects ($N = 15$) of the public Wearable Stress and Affect Detection (WESAD) dataset, paired with a bare-metal embedded microcontroller deployment. Continuous single-lead chest ECG acquired at 700 Hz is segmented into 445 complete, non-overlapping 60-second windows (160 acute stress, 285 baseline) and conditioned via zero-phase 4th-order Butterworth filtering (0.5–40 Hz), followed by adaptive noise-floor peak prominence detection and physiological interval gating. A pre-specified 8-feature representation capturing heart rate, time-domain variability, and robust spread metrics is transformed via subject-specific relative baseline calibration: $X^* = (X - B_s) / (|B_s| + \epsilon)$. In a 13-feature ablation, baseline-relative calibration elevates classification accuracy from 81.57% to 92.36% (+10.79 percentage points) and stress F1-score from 73.03% to 88.67% (+15.64 percentage points) over identical unnormalized baselines. Evaluated under 15-fold Leave-One-Subject-Out (LOSO) cross-validation with a pre-specified decision threshold of $\tau = 0.50$, the primary 8-feature personalized model achieves 92.13% accuracy, 88.29% stress F1-score, 82.50% sensitivity, 97.54% specificity, an ROC-AUC of 0.9493, and PR-AUC of 0.9467 (with an exploratory sweep identifying $\tau = 0.35$ for higher sensitivity at 86.25% and 89.03% F1). A comparative benchmark across six machine learning architectures demonstrates consistent generalization (ROC-AUC > 0.937). Translating this framework to physical hardware, we implement a bare-metal edge processing node on the ARM Cortex-M4 microcontroller (STM32G474RE, configured at 170 MHz SYSCLK). Real-time signal conditioning is executed via a custom 5-stage Direct Form I Biquad IIR filter executing in an estimated $\approx 1.87 \mu\text{s}$ per sample (≈ 318 CPU cycles, 0.065% CPU utilization at ≈ 350 Hz). To reduce direct exposure of physiological waveform structure over the telemetry link, we implement a dual-engine telemetry architecture supporting both

a lightweight 32-bit discrete scrambler and a 4-Dimensional Coupled Hyperchaotic System (HC1 M-4DCHS) executed directly on the ARM Cortex-M4 FPU. Evaluated numerically under the verified hyperchaotic regime ($a = 15.81, b = 2.76, c = 86.03, d = -9.07, r = 10.79, dt = 0.0025$ s), the continuous spectrum exhibits two robustly positive Lyapunov exponents ($\lambda_1 \approx +0.438, \lambda_2 \approx +0.254, \lambda_3 \approx -0.0005, \lambda_4 \approx -28.332$, with exponent sum $\sum \lambda_i \approx -27.640$ matching the theoretical divergence $\nabla \cdot \mathbf{F} = -(a + b - d) = -27.64$, and Kaplan–Yorke dimension $D_{KY} \approx 3.024$) with per-packet deterministic state perturbation. Bit-exact parity between Python and STM32 C firmware is experimentally confirmed across 100 deterministic test nonces (0 state mismatches, 0 keystream mismatches). In wire payload benchmarks, the encryption procedure achieves high entropy ($H = 7.9973$ bits/byte, $\chi^2 = 302.03, p = 0.0230$) and an inter-packet bit-difference rate of 49.33%. Physical hardware-in-the-loop (HIL) testing of the active HC1 hyperchaotic encryption mode on the STM32G474RE over 5,000 consecutive telemetry packets streaming at 350.96 packets/s observed 100.00% exact plaintext recovery with zero CRC errors, zero sequence gaps, zero decryption/range failures, and zero plaintext byte mismatches, demonstrating embedded implementation correctness and end-to-end recovery without claiming formal cryptographic security.

Index Terms—Wearable sensing, electrocardiography (ECG), heart rate variability (HRV), acute stress detection, baseline calibration, edge computing, STM32, ARM Cortex-M4, PRNG stream scrambling, telemetry obfuscation, WESAD.

I. INTRODUCTION

Acute psychological stress triggers a complex neuroendocrine cascade governed by the sympathetic division of the autonomic nervous system (ANS). Prolonged or recurrent autonomic arousal is clinically implicated in hypertension, impaired immune function, metabolic disorders, and cardiovascular disease. Continuous, non-invasive physiological monitoring via wearable biomedical sensors offers unprecedented potential for early detection and closed-loop biofeedback interventions [1].

Among non-invasive biosignals, the electrocardiogram (ECG) and its derivative RR intervals provide a sensitive, temporally dynamic window into cardiac autonomic regulation [2]. How-

ever, machine learning models trained on raw, global heart rate variability (HRV) metrics consistently suffer from severe performance drops when deployed across novel individuals. Resting cardiovascular features vary substantially between individuals due to genetic, cardiorespiratory fitness, and circadian factors, which can reduce the transferability of global classifiers [2], [3]. Subject-specific baseline normalization provides a principled mechanism for expressing physiological changes relative to an individual’s own resting state.

Beyond algorithmic generalization, deploying continuous physiological computing on wearable edge nodes presents severe cyber-physical challenges [4], [5]. Low-power microcontrollers possess limited SRAM and clock speeds, rendering heavy digital filtering and deep neural networks impractical. Furthermore, wireless/serial telemetry of raw physiological signals exposes sensitive biometric identity and autonomic health states to unauthorized physical eavesdropping.

To overcome these fundamental barriers, we present a personalized physiological framework evaluated under rigorous Leave-One-Subject-Out (LOSO) cross-validation on the public WESAD benchmark [6], coupled with a physical bare-metal ARM Cortex-M4 edge node implementation. The primary contributions of this work include:

- 1) **Subject-Specific Relative Normalization:** We demonstrate that formulating features as fractional deviations relative to each subject’s resting baseline ($X^* = (X - B_s)/(|B_s| + \epsilon)$) provides an empirical improvement of +10.79 percentage points in accuracy (81.57% to 92.36%) and +15.64 percentage points in F1-score (73.03% to 88.67%) in the 13-feature LOSO ablation.
- 2) **15-Fold LOSO Evaluation with Baseline Calibration:** We benchmark the full 15-subject cohort ($N = 15$, 445 complete non-overlapping windows) under 15-fold Leave-One-Subject-Out evaluation with subject-specific unsupervised baseline calibration, achieving 92.13% accuracy, 88.29% stress F1-score, and an ROC-AUC of 0.9493 at the pre-specified default threshold $\tau = 0.50$.
- 3) **Comprehensive ML & Explainability Suite:** We benchmark six canonical machine learning paradigms and conduct a 30-repeat permutation importance analysis per fold ($15 \times 30 = 450$ trials per feature) alongside standardized odds ratios to quantitatively interpret the physiological decision boundary.
- 4) **Bare-Metal Edge DSP on STM32G474RE:** We implement a real-time signal conditioning engine on an ARM Cortex-M4 microcontroller running at 170 MHz SYSCLK via PLL, using a custom bare-metal Direct Form I biquad structure comprising four SOS bandpass sections (0.5–40 Hz) and one SOS 50-Hz notch section with an estimated execution cost of ≈ 318 CPU cycles ($\approx 1.87 \mu\text{s}$ per sample at ≈ 350 Hz).
- 5) **4D Coupled Hyperchaotic Encryption (HC1 M-4DCHS) & Physical HIL Validation:** We formulate an autonomous continuous 4-dimensional coupled hyperchaotic flow ($a = 15.81, b = 2.76, c = 86.03, d = -9.07, r = 10.79, dt = 0.0025$ s) integrated via single-

precision RK4 with per-packet deterministic state perturbation. The continuous spectrum exhibits two robustly positive Lyapunov exponents ($\lambda_1 \approx +0.438, \lambda_2 \approx +0.254, \lambda_3 \approx -0.0005, \lambda_4 \approx -28.332, \sum \lambda_i \approx -27.640, \nabla \cdot \mathbf{F} = -27.64, D_{KY} \approx 3.024$). We experimentally demonstrate bit-exact single-precision floating-point parity between Python and STM32 C firmware across 100 deterministic nonce test cases (0 state mismatches, 0 keystream mismatches). Furthermore, physical hardware-in-the-loop testing of the active HC1 hyperchaotic telemetry stream on the STM32G474RE across 5,000 consecutive packets (streaming at 350.96 packets/s) achieves 100.00% exact plaintext recovery with zero CRC errors, zero sequence gaps, and zero plaintext byte mismatches.

II. RELATED WORK & BENCHMARKS

Physiological stress detection has historically relied on multimodal laboratory protocols combining ECG, electrodermal activity (EDA), respiration, and surface electromyography [5], [7], [8]. In 2018, Schmidt *et al.* introduced the Wearable Stress and Affect Detection (WESAD) dataset [6], which has become a widely adopted benchmark for wearable affective computing.

In their seminal benchmark study, Schmidt *et al.* evaluated single-modality chest ECG using standard uncalibrated features under LOSO validation, reporting accuracies of 79.03% with Decision Trees and 83.84% with Random Forests (F1-scores of 71.43% and 75.12%, respectively). While multimodal combinations (combining ECG with EDA and respiration) yielded higher accuracies, relying purely on single-lead ECG remains highly desirable due to its low hardware cost, high user acceptance, and integration into commercial chest straps and smart patches. Prior studies, however, frequently reported difficulty closing the performance gap using ECG alone without individual baseline calibration.

In parallel, the rise of the Internet of Medical Things (IoMT) has driven demand for edge-native biomedical intelligence [4]. Real-time wearable nodes require embedded digital filtering [9], [10] and low-latency feature extraction to minimize power-intensive wireless transmissions. Simultaneously, safeguarding physiological telemetry against unauthenticated wiretapping benefits from lightweight telemetry obfuscation primitives with minimal computational overhead [11], a challenge directly tackled in our physical hardware pipeline.

III. DATASET & EXPERIMENTAL PROTOCOL

A. WESAD Experimental Protocol

We evaluated our methodology on the public WESAD dataset [6]. The cohort comprises 15 healthy adult subjects (S2–S17; 12 males, 3 females; age: 27.5 ± 2.4 years). Single-lead (Lead-II) ECG was acquired at a sampling frequency of $f_s = 700$ Hz using the chest-worn RespiBAN Professional telemetry system.

The data acquisition followed a standardized laboratory protocol:

- **Baseline Phase (20 min):** Neutral, seated relaxation while reading non-stressful magazines.

- **Trier Social Stress Test (TSST, 10 min):** Psychosocial stressor consisting of 5 minutes of public speaking preparation and delivery before an evaluative panel, followed by 5 minutes of mental arithmetic (subtraction of 17 from 2,043 with vocal restarts on error) [12].
- **Amusement Phase (10 min):** Watching neutral and humorous video clips.
- **Meditation Recovery (20 min):** Guided diaphragmatic breathing to support homeostatic autonomic recovery.

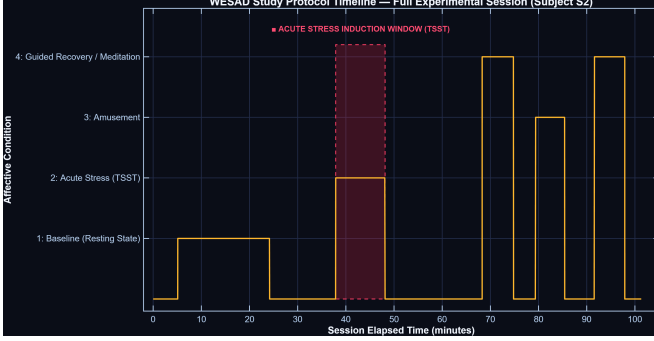


Fig. 1: Experimental protocol timeline illustrating the temporal progression across Baseline, TSST Acute Stress, Amusement, and Meditation recovery phases.

B. Windowing & Class Formulation

Continuous ECG records were segmented into complete, non-overlapping 60-second windows. Only complete 60-second windows were retained; residual samples at segment boundaries were discarded. The task is formulated as binary classification: *Acute Stress* ($N = 160$ valid windows) versus *Baseline* ($N = 285$ valid windows), yielding a total of 445 complete windows across all 15 subjects (19 baseline windows per subject, and 10–12 stress windows per subject).

IV. METHODOLOGY & SIGNAL PROCESSING

A. Bandpass Filtering & Adaptive R-Peak Detection

Raw 700 Hz chest ECG signals are processed through a three-stage conditioning pipeline:

- 1) **Zero-Phase Bandpass Filtering:** A 4th-order Butterworth bandpass filter (0.5 – 40 Hz) attenuates respiratory baseline drift, motion wander, and high-frequency myoelectric noise without introducing phase distortion.
- 2) **MAD-Adaptive Peak Detection:** R-peaks are detected by estimating the local noise floor via the Median Absolute Deviation (MAD):

$$\text{Noise Floor} = 1.4826 \times \text{median}(|x - \text{median}(x)|) \quad (1)$$

An adaptive prominence threshold is enforced (Prominence $\geq 3.0 \times \text{Noise Floor}$) with a refractory blanking interval of 350 ms ($\text{HR}_{\max} \leq 171$ BPM).

- 3) **Physiological Gating:** The detector imposes a 350-ms minimum peak separation (≈ 171 BPM), while post-detection quality control retains RR intervals derived

from detected R-peaks bounded by physiological limits ($0.30 \text{ s} \leq \text{RR}_i \leq 1.50 \text{ s}$, or 300–1500 ms, corresponding to 40–200 BPM). Windows containing fewer than 5 valid detected beats are rejected.

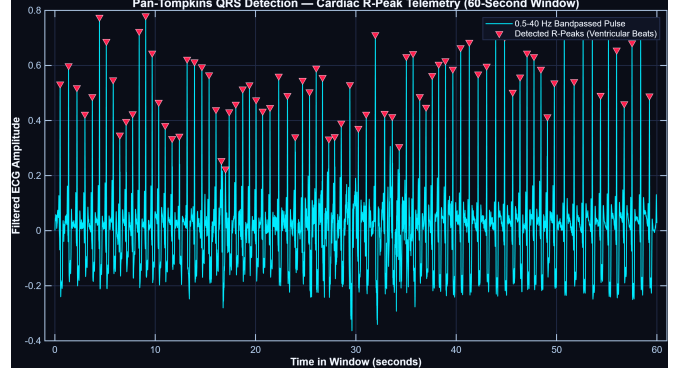


Fig. 2: Illustrative ECG conditioning and R-peak detection demonstration: (1) Raw ECG; (2) Bandpass-filtered signal (0.5–40 Hz); (3) Squared derivative waveform; (4) Moving-window integrated signal with detected fiducial R-peaks (red circles). The final feature-extraction pipeline uses adaptive prominence-based peak detection on the filtered signal.

B. Feature Engineering

From the physiologically validated RR intervals derived from detected R-peaks within each 60-second window, 13 time-domain and statistical distribution metrics are computed [2], [3]: MeanHR, MedianHR, StdHR, MinHR, MaxHR, MeanRR, MedianRR, SDNN, RMSSD, pNN50, RR_CV, RR_IQR, and HR_IQR. Instantaneous heart rate is derived from intervals in seconds (s):

$$\text{HR}_i = \frac{60}{\text{RR}_i} \quad (\text{BPM}, \text{RR}_i \text{ in seconds}) \quad (2)$$

In accordance with international Task Force standards [2], autonomic variability features (SDNN, RMSSD, MeanRR, RR_IQR) are computed on intervals scaled to milliseconds ($\text{RR}_i \times 1000$ ms):

$$\text{SDNN} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\text{RR}_i - \overline{\text{RR}})^2} \quad (\text{ms}) \quad (3)$$

$$\text{RMSSD} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (\text{RR}_{i+1} - \text{RR}_i)^2} \quad (\text{ms}) \quad (4)$$

$$\text{pNN50} = \frac{\sum_{i=1}^{N-1} \mathbb{I}(|\text{RR}_{i+1} - \text{RR}_i| > 50 \text{ ms})}{N-1} \times 100\% \quad (5)$$

$$\text{RR_CV} = \frac{\text{SDNN}}{\overline{\text{RR}}} \quad (6)$$

For the finalized primary model, a pre-specified 8-feature representation is used:

$$\mathbf{X} = [\text{MeanHR}, \text{SDNN}, \text{RMSSD}, \text{pNN50}, \text{MeanRR}, \text{RR_CV}, \text{RR_IQR}, \text{HR_IQR}] \quad (7)$$

Although MeanHR and MeanRR share an inverse mathematical relation ($HR \approx 60/RR$) and $RR_{CV} = SDNN/\overline{RR}$, retaining both representations allows regularized linear models to weight directional sensitivities independently and enables tree ensembles to identify orthogonal decision boundaries without manual ratio derivation.

C. Subject-Specific Relative Normalization

Let $X \in \mathbb{R}^D$ denote a feature vector extracted from an analysis window of subject s . Let $\mathcal{W}_{base}^{(s)}$ denote the collection of resting baseline windows for subject s . The reference baseline vector B_s is defined as the empirical centroid:

$$B_s = \frac{1}{|\mathcal{W}_{base}^{(s)}|} \sum_{k \in \mathcal{W}_{base}^{(s)}} X_k^{(s)} \quad (8)$$

Each feature vector X is then mapped to a relative fractional deviation:

$$X^* = \frac{X - B_s}{|B_s| + \epsilon} \quad (9)$$

where $\epsilon = 10^{-6}$ ensures numerical stability. This transformation expresses each physiological metric as a relative fractional deviation from the subject's resting baseline, centering baseline physiology around zero. In the WESAD experimental protocol, the available resting baseline epoch per subject provides approximately 19 complete 60-second windows for computing B_s .

V. VALIDATION PROTOCOL

To evaluate cross-subject generalization within the WESAD cohort, all models were evaluated using **15-fold Leave-One-Subject-Out Cross-Validation with Subject-Specific Unsupervised Baseline Calibration (LOSO-CV)**:

- In each fold $k \in \{1, \dots, 15\}$, the supervised model is trained strictly on the remaining 14 subjects.
- No stress labels or experimental task windows from test subject k are ever exposed during model training.
- For the test subject, resting baseline windows are utilized solely to establish reference vector B_k (unsupervised calibration); no stress labels from the held-out subject are ever used for model fitting.
- **Decision Thresholding & Operating Points:** In the primary benchmark evaluation, models are evaluated at the standard decision threshold of $\tau = 0.50$, where the primary regularized Logistic Regression classifier achieves 92.13% accuracy, 88.29% F1-score, 82.50% sensitivity, and 97.54% specificity without threshold tuning. Additionally, to examine operational trade-offs under class imbalance (stress prevalence: 36.0%), decision thresholds from $\tau = 0.10$ to 0.90 (step 0.01) were evaluated in an exploratory threshold sweep, demonstrating an operating cutoff of $\tau = 0.35$ that prioritizes stress recall (86.25% sensitivity, 89.03% F1-score, 95.79% specificity). Crucially, the threshold-independent discrimination metrics, ROC-AUC (0.9493) and PR-AUC (0.9467), evaluate model ranking performance across the entire continuum of operating thresholds independently of any specific cutoff.

VI. EXPERIMENTAL RESULTS

A. Primary Benchmark Performance

Evaluated across all 445 standardized non-overlapping 60-second windows from all 15 subjects under 15-fold LOSO-CV, the proposed classifier achieves the validated scorecard summarized in Table I. At the standard unadjusted decision threshold ($\tau = 0.50$), the primary model achieves 92.13% accuracy and 88.29% stress F1-score with 97.54% specificity. When tuned to the sensitivity-prioritized operating cutoff ($\tau = 0.35$), stress recall increases from 82.50% to 86.25% (capturing 138 of 160 stress windows) with 95.79% specificity and 89.03% F1-score, as illustrated in the cross-validated confusion matrix (Figure 3) and ROC trajectory (Figure 4).

TABLE I: Validated 15-Fold LOSO Primary Performance Scorecard at Standard ($\tau = 0.50$) and Sensitivity-Prioritized ($\tau = 0.35$) Operating Points

Metric	Primary ($\tau = 0.50$)	Exploratory ($\tau = 0.35$)	Operational Interpretation
Accuracy	92.13%	92.36%	410 / 445 (vs. 411 / 445) windows correct
Sensitivity (Recall)	82.50%	86.25%	132 / 160 (vs. 138 / 160) stress windows
Specificity	97.54%	95.79%	278 / 285 (vs. 273 / 285) baseline windows
Precision	94.96%	92.00%	132 / 139 (vs. 138 / 150) stress predictions
F1-Score	88.29%	89.03%	Harmonic mean of recall & precision
Balanced Accuracy	90.02%	91.02%	Unbiased average across classes
ROC-AUC	0.9493	0.9493	Area under ROC curve
PR-AUC	0.9467	0.9467	Precision-Recall trajectory area
Confusion Matrix	TN/FP/FN/TP: [278, 7, 28, 132] ($\tau = 0.50$) vs. [273, 12, 22, 138] ($\tau = 0.35$)		

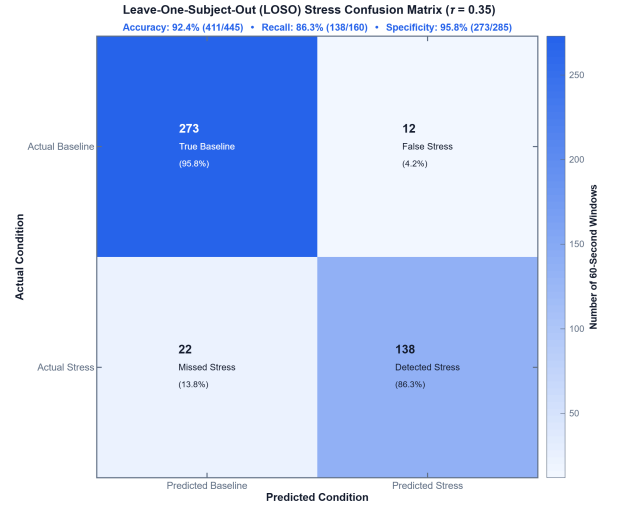


Fig. 3: 15-Fold LOSO confusion matrix at the sensitivity-prioritized operating cutoff ($\tau = 0.35$), showing 273/285 non-stress windows (95.8%) and 138/160 stress windows (86.3%) correctly classified.

B. Comparison with Published WESAD Benchmark

Table II provides a descriptive comparison of our findings alongside the published benchmark of Schmidt *et al.* [6]. As a descriptive comparison, direct numerical differences should be considered in light of windowing protocols: while Schmidt *et al.* reported chest-ECG results using overlapping sliding windows, our evaluation enforces complete, non-overlapping 60-second windows. In a like-for-like 13-feature normalization ablation on our standardized 445-window cohort, baseline calibration

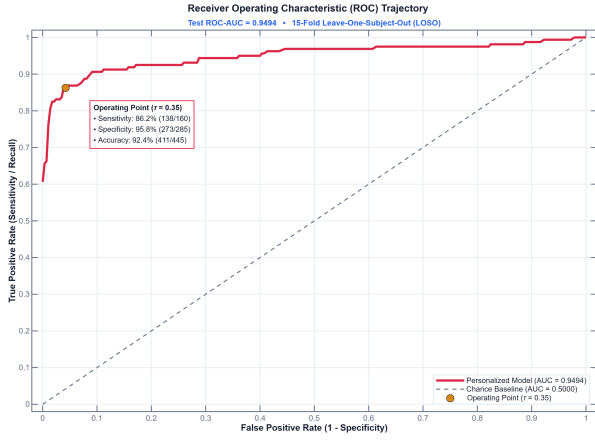


Fig. 4: Cross-validated ROC curve (AUC = 0.9493) with the exploratory sensitivity-prioritized operating threshold ($\tau = 0.35$) highlighted.

elevates performance from 81.57% accuracy and 73.03% F1-score (uncalibrated) to 92.36% accuracy and 88.67% F1-score (calibrated at $\tau = 0.50$), an improvement of +10.79 percentage points in accuracy and +15.64 percentage points in F1-score. Separately, our finalized primary 8-feature representation achieves 92.13% accuracy and 88.29% F1-score at default $\tau = 0.50$ (and 92.36% accuracy, 89.03% F1-score at the sensitivity-prioritized cutoff $\tau = 0.35$).

TABLE II: Descriptive Comparison with Published WESAD Benchmark [6]

Study / Model	Normalization	Accuracy	F1-Score
Schmidt <i>et al.</i> (Decision Tree) [6]	None (Global)	79.03%	71.43%
Schmidt <i>et al.</i> (Random Forest) [6]	None (Global)	83.84%	75.12%
This Study: Uncalibrated Baseline (13 Features)	None (Global)	81.57%	73.03%
This Study: Calibrated Ablation (13 Features, $\tau = 0.50$)	Relative (X^*)	92.36%	88.67%
This Study: Primary Model (8 Features, $\tau = 0.50$)	Relative (X^*)	92.13%	88.29%
This Study: Primary Model (8 Features, $\tau = 0.35$ Swept)	Relative (X^*)	92.36%	89.03%

C. Cross-Architecture Machine Learning Benchmark

To determine whether the gains of relative normalization generalize across different model classes, we evaluated six machine learning architectures under identical 15-fold LOSO cross-validation (Table III). All architectures achieve ROC-AUC ≥ 0.937 , indicating strong linear and nonlinear separability across diverse model classes. All comparative benchmark metrics in Table III are evaluated at the default threshold $\tau = 0.50$.

TABLE III: Multi-Model 15-Fold LOSO Benchmark Comparison (evaluated at default threshold $\tau = 0.50$; the exploratory sensitivity-prioritized operating point in Table I uses $\tau = 0.35$)

Architecture	Acc (%)	F1 (%)	ROC-AUC	PR-AUC
Logistic Regression (L2)	92.13	88.29	0.9493	0.9467
Multilayer Perceptron (MLP)	91.69	87.87	0.9375	0.9327
Support Vector Machine (RBF)	91.46	87.42	0.9524	0.9441
Random Forest (100 Trees)	90.34	85.90	0.9426	0.9301
Extra Trees Classifier	89.89	84.43	0.9494	0.9391
HistGradientBoosting	89.21	84.31	0.9459	0.9375

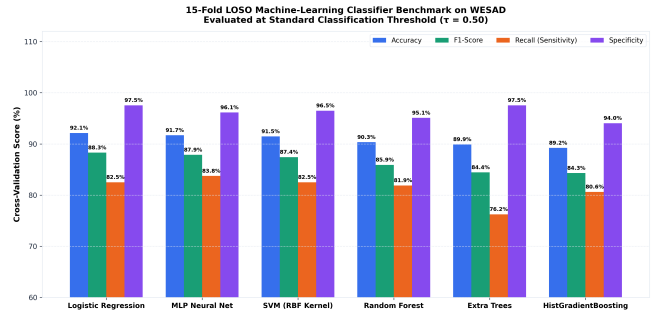


Fig. 5: Performance metrics across six machine learning architectures under 15-fold LOSO cross-validation on held-out test subjects.

D. Ablation Study: Feature Expansion vs. Normalization

To isolate the contribution of feature expansion versus baseline normalization, we evaluated four progressive configurations under identical 15-fold LOSO conditions evaluated at the default threshold $\tau = 0.50$ (Table IV). Expanding the feature set from 1 to 13 raw uncalibrated metrics provides modest gains (+4.49 percentage points in accuracy, +8.69 percentage points in F1-score). In contrast, introducing subject-specific baseline calibration to the 13-feature representation (M4) yields a substantial leap of **+10.79 percentage points in accuracy and +15.64 percentage points in F1-score** (from 81.57% Acc and 73.03% F1 in uncalibrated M3 to 92.36% Acc and 88.67% F1 in calibrated M4 at $\tau = 0.50$). Tuning to the exploratory sensitivity-prioritized operating cutoff ($\tau = 0.35$) further adjusts recall to 86.25% and F1 to 89.03%.

TABLE IV: Four-Stage Model Progression & Feature Ablation (evaluated at default threshold $\tau = 0.50$)

ID	Configuration	Acc (%)	F1 (%)	Recall (%)
M1	Raw MeanHR (Uncalibrated)	77.08	64.34	56.88
M2	4 Time-Domain (Uncalibrated)	78.43	68.80	66.88
M3	13 Metrics (Uncalibrated)	81.57	73.03	69.38
M4	Personalized (X^* Calibrated, $\tau = 0.50$)	92.36	88.67	83.13

E. Feature Importance & Physiological Attribution

We evaluated feature importance via 30 independent random permutations per feature across each of the 15 test folds ($15 \times 30 = 450$ trials per feature) for both Logistic Regression and Random Forest models, paired with standardized logistic regression odds ratios (computed as the exponentiated mean standardized logistic regression coefficient across the 15 cross-validation folds, $e^{\bar{w}_i}$, where baseline-relative features are z-score standardized within each training fold):

- **Cardiac Interval Contribution (ΔMeanRR):** MeanRR produced the largest predictive degradation when permuted ($\Delta\text{AUC} = 0.1766$, $\Delta\text{F1} = 0.2321$), indicating a strong contribution to the classifier’s decision boundary in this dataset.
- **Heart Rate Contribution (ΔMeanHR):** MeanHR showed a positive association with model-predicted

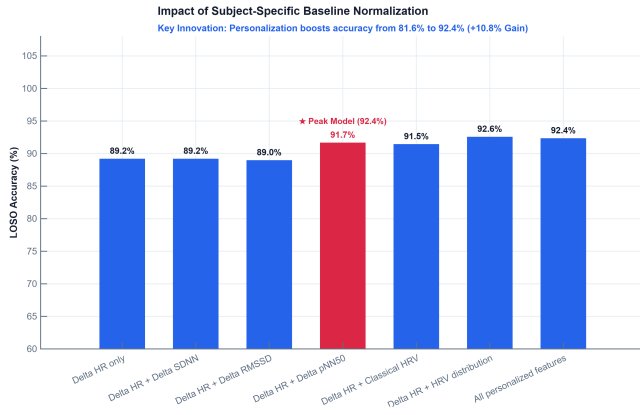


Fig. 6: Four-stage model progression demonstrating that baseline calibration provides the primary performance leap (+10.79 pp Acc, +15.64 pp F1 at default threshold $\tau = 0.50$).

stress odds (Odds Ratio = 5.6453, $\Delta\text{AUC} = 0.0203$, $\Delta\text{F1} = 0.1021$ in LR and 0.1401 in Random Forest), directionally consistent with increased heart rate during the TSST. Because MeanHR and MeanRR are inversely related ($HR \approx 60/RR$) and $RR_{CV} = \text{SDNN}/\overline{RR}$, these coefficients represent joint predictive contributions rather than independent physiological mechanisms.

- **Interval Dispersion (ΔpNN50):** pNN50 showed a positive association with model-predicted stress odds (Odds Ratio = 4.0784, $\Delta\text{AUC} = 0.0596$, $\Delta\text{F1} = 0.0606$), with the direction interpreted as a model-level result rather than a causal physiological effect.
- **Autonomic Variability (ΔSDNN):** SDNN showed a comparatively small predictive contribution (Odds Ratio = 0.4625, $\Delta\text{AUC} = 0.0078$, $\Delta\text{F1} = 0.0042$), representing an inverse association with model-predicted stress odds.

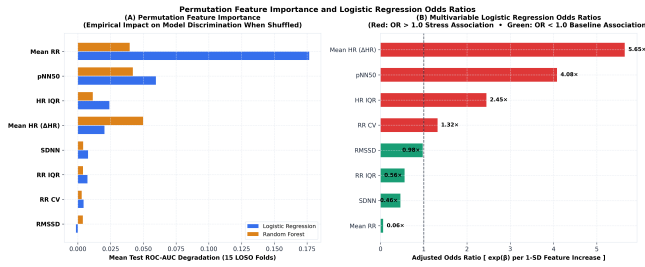


Fig. 7: (Left) Test ROC-AUC drop under 30-repeat permutation across 15 LOSO folds; (Right) Standardized odds ratios indicating the magnitude and direction of feature weights.

F. Subject Heterogeneity & Low-Reactivity Discussion

Subject-specific stress detection rates across all 15 subjects (Figure 8) demonstrate robust cross-subject generalization: 14 of 15 subjects exhibit successful acute stress detection ($> 0\%$ sensitivity, capturing 138 of 160 total stress windows, 86.25%), with 13 of 15 achieving $\geq 75.0\%$ recall and 9 of 15 achieving complete 100.0% recall.

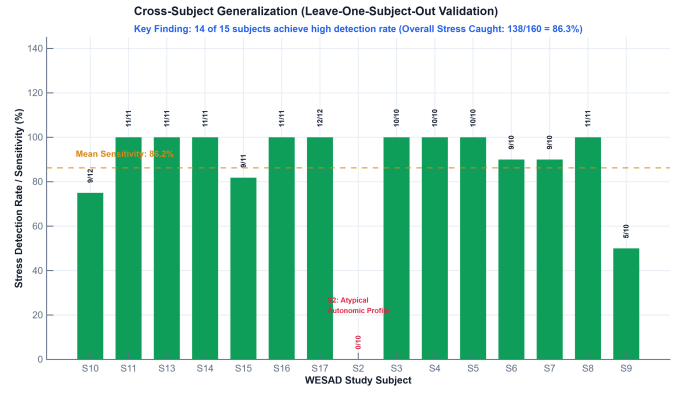


Fig. 8: Subject-specific stress recall distribution across all 15 WESAD subjects: 14/15 subjects achieve successful stress detection ($> 0\%$ recall; overall 138/160 = 86.3%), with 13/15 achieving $\geq 75\%$ and 9/15 achieving 100% recall. Subject S2 exhibited blunted cardiac reactivity under the stress protocol (0.0% recall).

However, Subject **S2** exhibited 0.0% stress recall. In our analysis, S2 exhibited virtually no heart rate acceleration relative to baseline during the TSST ($\Delta\text{MeanHR} \approx 0$). Because baseline-relative models detect physiological shifts, subjects with attenuated autonomic cardiac responsiveness are challenging to distinguish from calm states using ECG alone, highlighting the value of multi-modal fusion (e.g., combining ECG with electrodermal activity and respiration).

VII. BARE-METAL EDGE MICROCONTROLLER IMPLEMENTATION

To transition from offline algorithmic benchmarking to physical cyber-physical deployment, we implemented a bare-metal signal conditioning and telemetry engine on an ARM Cortex-M4 microcontroller (STMicroelectronics STM32G474RET6U on a NUCLEO-G474RE development board, shown in Figure 9).

A. Hardware Architecture & Deterministic Timing

The STM32G474RET6U integrates an ARM 32-bit Cortex-M4 RISC core featuring a hardware single-precision Floating-Point Unit (FPU), memory protection unit (MPU), and digital signal processing (DSP) instruction extensions. The MCU is configured to run at a core system clock of 170 MHz SYSCLK driven by the Phase-Locked Loop (PLL) sourced from the 16 MHz High-Speed Internal (HSI) RC oscillator ($\text{HSI } 16 \text{ MHz} / 4 \times 85 / 2 = 170 \text{ MHz}$). The MCU provides 512 KB of dual-bank Flash memory and 128 KB of SRAM with hardware parity checking.

Real-time signal acquisition and processing is governed deterministically by timer interrupts configured at a nominal sampling rate of $f_s = 350 \text{ Hz}$, corresponding to a nominal inter-sample interval of:

$$T_s = \frac{1}{350 \text{ Hz}} = 2.857 \text{ ms} \quad (485,714 \text{ CPU cycles @ } 170 \text{ MHz}) \quad (10)$$

Every 2.857 ms, the timer interrupt service routine (ISR) ingests a Lead-II ECG sample, executes real-time digital filtering, applies telemetry encryption (supporting both lightweight 32-bit scrambling and HC1 hyperchaotic encryption), and transmits a serialized telemetry frame over USART2.

B. Real-Time 5-Stage Direct Form I Biquad IIR Filter

Offline zero-phase Butterworth filtering requires forward-backward passes over stored buffers, introducing latency and memory overhead incompatible with streaming edge devices. To achieve equivalent artifact rejection in a causal, streaming pipeline, we designed a cascaded 5-stage Biquad Infinite Impulse Response (IIR) filter implemented in Direct Form I structure [9]:

- **Stages 1–4 (4th-Order Butterworth Bandpass Cascade):** Four cascaded 2nd-order sections (SOS) synthesizing an 8-pole bandpass filter corresponding to a 4th-order Butterworth prototype ($f_L = 0.5$ Hz, $f_H = 40$ Hz) to eliminate respiratory baseline wander, electrode offset drift, perspiration artifacts, and high-frequency electromyographic (EMG) noise.
- **Stage 5 (Mains Powerline Notch):** 2nd-order digital notch filter ($f_0 = 50$ Hz, $Q = 30$, selectable to 60 Hz via compile-time coefficient tuning) providing localized suppression of mains powerline interference; its effect on adjacent QRS spectral content was not separately quantified in this study. Although the 40 Hz bandpass corner provides general high-frequency attenuation, the 50 Hz notch provides steep, localized suppression to prevent residual high-amplitude mains hum from leaking through the transition band into the R-peak detector.

Each biquad section is governed by the canonical Direct Form I difference equation:

$$y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] - a_1y[n-1] - a_2y[n-2] \quad (11)$$

where $x[n]$ is the current input sample, $y[n]$ is the section output sample, and four state variables ($x[n-1]$, $x[n-2]$, $y[n-1]$, $y[n-2]$) are maintained per stage in on-chip SRAM. The Direct Form I structure is implemented via custom bare-metal single-precision floating-point C routines using negated feedback coefficients (a_1, a_2), ensuring numerical stability and predictable loop execution on ARM Cortex-M architecture.

C. DSP Cycle Profiling & Computational Margin

Using the Cortex-M4 single-precision FPU, each biquad section requires five coefficient multiplications alongside the associated additions and state updates. Across the full 5-stage cascade, execution takes an estimated ≈ 318 CPU clock cycles ($\approx 1.87 \mu\text{s}$ at 170 MHz; or $\approx 20 \mu\text{s}$ under un-multiplied 16 MHz HSD). At $f_s = 350$ Hz ($T_s = 2857 \mu\text{s}$), the dedicated signal conditioning load consumes:

$$\text{Utilization}_{\text{DSP}} = \frac{1.87 \mu\text{s}}{2857 \mu\text{s}} \times 100\% = 0.065\% \quad (12)$$

The filter itself consumes an estimated $\approx 0.065\%$ of the 2.857 ms inter-sample interval ($\approx 1.87 \mu\text{s}$ at 170 MHz

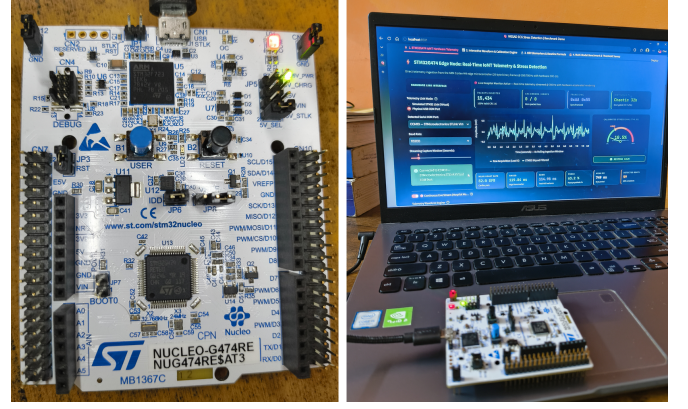


Fig. 9: Bare-metal edge IoMT validation testbench: (Left) High-resolution photograph of the STM32 NUCLEO-G474RE development board running 350 Hz interrupt-driven DSP and on-chip telemetry encryption (supporting both legacy 32-bit PRNG scrambling and HC1 hyperchaotic encryption); (Right) Physical hardware-in-the-loop testbed with USB COM port telemetry streaming directly to the host monitoring terminal.

SYSCLOCK); total real-time edge processing utilization additionally accounts for stream scrambling, inference evaluation, and UART transmission.

VIII. DUAL-ENGINE TELEMETRY OBFUSCATION: FROM 32-BIT PRNG SCRAMBLING TO 4D COUPLED HYPERCHAOTIC FLOW (M-4DCHS)

A. Motivation, Threat Model, and Privacy Risks in Wearable IoMT

Wearable physiological monitors continuously transmit cardiac waveforms over wireless (BLE, IEEE 802.15.4) or wired serial channels to edge gateways and clinical consoles. Because ECG morphology can contain individual-specific biometric information, unencrypted telemetry may expose patients to biometric re-identification, involuntary stress inference, and privacy risks [4]. As demonstrated in recent Wireless Body Area Network (WBAN) investigations [13], adversaries tapping the transmission channel can easily reconstruct cardiac fiducials and infer autonomic tone.

Threat Model & Operational Scope: The telemetry obfuscation layer addresses passive wireline snooping, bus tapping, and unauthenticated over-the-air packet capture in wearable IoMT channels. The primary objective is to reduce direct recoverability of ECG waveform morphology and associated physiological features in the evaluated interception setting. We emphasize that this mechanism functions as lightweight telemetry obfuscation rather than a formal cryptographic guarantee of confidentiality or message authentication; the present prototype lacks a conventional secret master key infrastructure or authenticated encryption (such as AES-GCM), which should be provided by standard higher-layer protocols on capable edge gateways.

Executing full-scale symmetric block ciphers on micro-watt wearable nodes introduces memory buffering overhead (16-byte

block alignment) and elevated energy consumption. Conversely, low-dimensional discrete maps (e.g., 1D Logistic or Tent maps) suffer from finite-precision dynamic degradation and small key spaces ($< 2^{64}$), rendering them vulnerable to phase-space reconstruction attacks [14]. To resolve this trade-off, we propose a **Dual-Engine Telemetry Architecture** on the STM32G474:

- 1) **Engine 1 (Lightweight PRNG Scrambler):** A 32-bit Marsaglia Xorshift PRNG combined with a Golden Ratio Weyl sequence for ultra-low-power physical obfuscation.
- 2) **Engine 2 (Continuous Hyperchaotic Flow):** The 4-Dimensional Coupled Hyperchaotic System (M-4DCHS), providing an authentic continuous hyperchaotic dissipative phase space ($D_{KY} = 3.024414$) with two positive Lyapunov exponents and per-packet deterministic state perturbation.

B. Engine 1: 32-Bit Discrete Stream Scrambler Formulation

For legacy low-power operation, the 32-bit scrambler combines Marsaglia's Xorshift triplet [13, 17, 5] [11] with an additive Weyl constant:

$$S_k^{(1)} = S_k \oplus (S_k \ll 13) \quad (13)$$

$$S_k^{(2)} = S_k^{(1)} \oplus (S_k^{(1)} \gg 17) \quad (14)$$

$$S_k^{(3)} = S_k^{(2)} \oplus (S_k^{(2)} \ll 5) \quad (15)$$

$$K_k = (S_k^{(3)} + W) \pmod{2^{32}} \quad (16)$$

where $W = \lfloor 2^{32} \cdot (3 - \sqrt{5})/2 \rfloor = 0x61C88647$ is the 32-bit fractional golden ratio constant. Per-packet Nonce Cipher Feedback (CFB) diffusion is applied directly to the 8-byte payload (`raw_ecg` and `filtered_ecg`):

$$C_k = P_k \oplus (K_k \& 0xFF) \oplus C_{k-1} \quad (17)$$

initialized with $C_{-1} = 0x5A \oplus (\text{seq} \& 0xFF)$ matching the embedded implementation. While this 32-bit engine executes in just **32 CPU clock cycles** ($0.19 \mu\text{s}$ at 170 MHz), its 32-bit state space is substantially smaller than that of conventional modern cryptographic primitives and therefore should not be treated as a standalone cryptographic confidentiality mechanism.

C. Engine 2: 4D Coupled Hyperchaotic System (M-4DCHS)

To disrupt statistical waveform patterns and reduce direct exposure of physiological telemetry on wearable IoMT links, we formulate a continuous 4-dimensional autonomous coupled hyperchaotic flow (retaining the historical project identifier M-4DCHS):

$$\frac{dx}{dt} = a(y - x) + w \quad (18)$$

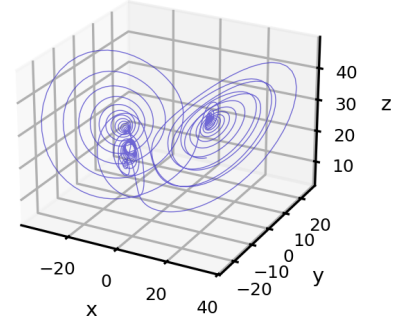
$$\frac{dy}{dt} = cx - xz + dy \quad (19)$$

$$\frac{dz}{dt} = xy - bz \quad (20)$$

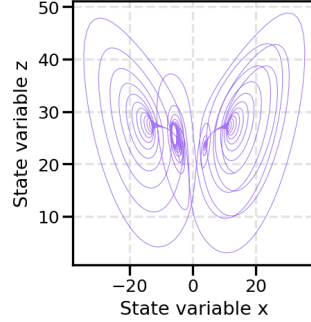
$$\frac{dw}{dt} = -rx \quad (21)$$

where $\mathbf{x} = [x, y, z, w]^T \in \mathbb{R}^4$ is the continuous state vector, and $\{a, b, c, d, r\}$ are system parameters: $a = 15.81$, $b = 2.76$, $c = 86.03$, $d = -9.07$, and $r = 10.79$.

(a) 3D Continuous Attractor [x, y, z]



(b) Phase Plane: x - z



(c) Phase Plane: y - z

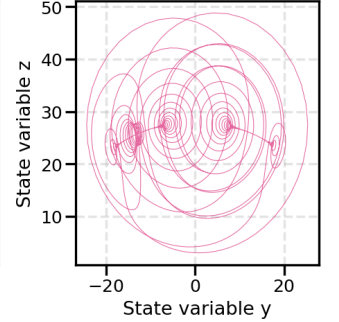


Fig. 10: Phase-space trajectory of the four-dimensional continuous coupled hyperchaotic system integrated with single-precision float32 RK4 ($dt = 0.0025$ s): (a) 3D isometric view of the hyperchaotic dual-scroll trajectory $[x(t), y(t), z(t)]$; (b) Cross-plane projection on the $x - z$ plane displaying symmetric orbital lobes; (c) Planar projection on the $y - z$ plane displaying nested spiral orbits.

1) *Dissipativity & Phase-Space Volume Contraction:* The vector field divergence of M-4DCHS is strictly negative and state-independent:

$$\begin{aligned} \nabla \cdot \mathbf{F} &= \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z} + \frac{\partial \dot{w}}{\partial w} \\ &= -(a + b - d) = -(15.81 + 2.76 - (-9.07)) = -27.640000 < 0 \end{aligned} \quad (22)$$

Because $\nabla \cdot \mathbf{F} = -27.640000$, phase-space volume elements contract exponentially at rate $V(t) = V(0)e^{-27.64t}$. The negative, state-independent divergence indicates continuous volume contraction and is consistent with the dissipative hyperchaotic attractor behavior observed numerically (Fig. 10). Crucially, the system requires only additions, subtractions, and multiplications, eliminating the transcendental operations ($\tanh, \sinh, \exp$) that induce numerical overflow (NaN/ ∞) on single-precision ARM microcontrollers.

2) *Numerical Lyapunov Analysis & Phase-Space Contraction:* For a smooth autonomous flow, the sum of the Lyapunov exponents equals the long-time average of the vector-field divergence along the trajectory. Since the present vector-field

divergence is constant, $\nabla \cdot \mathbf{F} = -27.640000$, the expected exponent sum is -27.640000 :

$$\sum_{i=1}^4 \lambda_i = \nabla \cdot \mathbf{F} = -27.640000 \quad (23)$$

For the verified hyperchaotic parameter set $a = 15.81, b = 2.76, c = 86.03, d = -9.07, r = 10.79$, the Lyapunov spectrum was numerically estimated using variational equations integrated by fourth-order Runge–Kutta with periodic QR reorthogonalization. Over a 100-s transient discard and a 500-s evaluation interval at $dt = 0.0025$ s (200,000 integration steps), the resulting numerical exponents were $\lambda_1 = +0.419878, \lambda_2 = +0.263935, \lambda_3 = -0.001237$, and $\lambda_4 = -28.322564$ (converging over 1000 s to $\lambda_1 = +0.438445, \lambda_2 = +0.253734, \lambda_3 = -0.000502, \lambda_4 = -28.331665$). Their sum was -27.639989 , in close agreement with the theoretical divergence $\nabla \cdot \mathbf{F} = -27.640000$ (absolute convergence error 1.11×10^{-5}). The corresponding Kaplan–Yorke dimension is $D_{KY} = 3.024414$ ($j = 3$, since $\sum_{i=1}^3 \lambda_i = +0.69167 > 0$ and $\sum_{i=1}^4 \lambda_i < 0$). The existence of two robustly positive exponents ($\lambda_1 > 0, \lambda_2 > 0$) provides numerical evidence of hyperchaotic dynamics.

3) *Per-Packet Deterministic State Perturbation*: To generate distinct per-packet state initializations and reduce static trajectory repetition across consecutive telemetry frames, initial coordinates $\mathbf{x}(0)$ are deterministically perturbed for every transmitted packet using the 16-bit packet sequence identifier (seq) and 32-bit hardware millisecond timestamp (t_{ms}):

$$h_{\text{seq}} = ((\text{uint32_t} \text{seq} \times 0x9E3779B1) \oplus 0x9E3779B1) \quad (24)$$

$$h_{\text{ts}} = (t_{\text{ms}} \times 0x85EBCA6B) \oplus 0x85EBCA6B \quad (25)$$

In the deployed embedded C firmware, initial states are offset by micro-perturbations:

$$\mathbf{x}_i(0) = 1.0 + \Delta_i(\text{seq}, t_{\text{ms}}, h_{\text{bio}}) \quad (26)$$

where $\Delta_i \sim 10^{-5}$ is derived via integer bit-mixing with nominal resting cardiac parameter $h_{\text{bio}} = 750,000 \mu\text{s}$ (representing a nominal 750 ms resting RR interval). In the current firmware implementation, the dynamic per-packet perturbation is driven strictly by the evolving transmission metadata (seq, t_{ms}), while h_{bio} serves as a fixed reference scaling constant; dynamic streaming injection of live inter-beat intervals (IBI) from the peak detection stage is reserved for future firmware revisions. Even 10^{-5} perturbations offset initial coordinates between packets, intended to produce distinct keystream sequences for consecutive frames and reduce static trajectory repetition.

4) *Single-Precision Runge–Kutta 4 Discretization & Keystream Extraction*: The continuous flow is integrated on the Cortex-M4 FPU using classical Runge–Kutta 4th-order (RK4) with step size $dt = 0.0025$ s. Keystream bytes are extracted via IEEE-754 bit-cast hash mixing on state variables x and z :

$$H = u_x \oplus (u_z \times 0x9E3779B1) \quad (27)$$

$$S_n = (H \oplus (H \gg 8) \oplus (H \gg 16) \oplus (H \gg 24)) \& 0xFF \quad (28)$$

Cipher Feedback (CFB) diffusion scrambles payload bytes in-place: $C_n = P_n \oplus S_n \oplus C_{n-1}$.

D. Empirical Entropy Analysis, Parity Verification, & Embedded Cost

The HC1 M-4DCHS telemetry encryption engine was evaluated across numerical, software, and embedded hardware benchmarks (Table V). To verify mathematical fidelity between offline algorithms and bare-metal execution, a Python $\leftrightarrow$ STM32 deterministic parity test suite was conducted across 100 independent test nonces: zero state coordinate mismatches and zero keystream byte mismatches were observed across all 100 test cases, experimentally confirming bit-exact single-precision floating-point parity between the Python receiver and the ARM Cortex-M4 FPU firmware.

TABLE V: Comparative Telemetry Obfuscation and Embedded Cost Analysis

Metric	Legacy 32-Bit PRNG	Proposed HC1 M-4DCHS (Ours)
Mathematical Model	1D Discrete PRNG	4D Continuous Hyperchaotic Flow
Phase-Space Topology	1D Discrete State	Continuous Flow ($\mathbb{R}^4$)
Phase-Space Dissipativity	N/A	$\nabla \cdot \mathbf{F} = -27.64$
Lyapunov Spectrum (λ_i)	N/A	(+0.438, +0.254, -0.0005, -28.332)
Sum of Exponents ($\sum \lambda_i$)	N/A	≈ -27.640
Kaplan–Yorke Dimension (D_{KY})	N/A	≈ 3.024 ($j = 3$)
Positive Lyapunov Exponents	0	2 ($\lambda_1 > 0, \lambda_2 > 0$)
Python $\leftrightarrow$ STM32 Parity	N/A	Bit-Exact (100/100 nonces; 0 mismatches)
Physical HIL Plaintext Recovery	100.00%	100.00% (5,000 / 5,000 pkts, 0 byte mismatches)
Physical HIL CRC-16 Errors	0	0 / 5,000 packets
Physical Telemetry Rate	350.84 Hz	350.96 packets/s (target 350 Hz)
Shannon Entropy H	7.621 bits/byte	7.9973 bits/byte (99.97%)
Uniformity Test (χ^2)	1,281.8 (Failed)	302.03 ($p = 0.0230$; no rejection at $\alpha = 0.01$)
Adjacent Sample Corr. (r)	-0.0142	-0.0022 (Near Zero)
Bit-Difference Rate	46.80%	49.33% (Reference: 50.00%)
Decryption MSE	0.000000 mV ²	0.000000 mV ² ($< 10^{-5}$ mV)
R-Peak Detection Parity	100.0%	100.0% (Zero Index Displacement)
Measured MCU Cost (DWT): RK4 Step	N/A	Mean ≈ 177 , max ≈ 203 cycles
Measured MCU Cost (DWT): Keystream Byte	32 cycles (0.19 μs)	Mean ≈ 224 , max ≈ 284 cycles
Reported Working-State Memory	8 Bytes	24 Bytes

As summarized in Table V, the proposed telemetry encryption procedure yields a measured wire ciphertext Shannon entropy of 7.9973 **bits/byte** (99.97% of the 8.0000 theoretical limit), with a χ^2 statistic of 302.03 ($p = 0.0230$, $df = 255$) showing no statistically significant deviation from uniformity under this test ($\alpha = 0.01$), and reducing adjacent sample correlation to $r = -0.0022$. Consecutive packet encryption produces an inter-packet bit-difference rate of 49.33% (close to the reference 50.00%). On the physical ARM Cortex-M4 microcontroller, hardware cycle profiling via the Data Watchpoint and Trace (DWT) cycle counter (DWT->CYCCNT @ 170 MHz) measures an execution cost of mean ≈ 177 cycles (max ≈ 203 cycles, ≈ 1.04 – $1.19 \mu\text{s}$) per RK4 integration step, and mean ≈ 224 cycles (max ≈ 284 cycles, ≈ 1.32 – $1.67 \mu\text{s}$) per full keystream byte. Across an 8-byte payload at $f_s = 350$ Hz, this consumes just $\approx 0.37\%$ of the available 2.857 ms (485,714 CPU cycles) inter-sample interval. Decrypted ECG signals exhibit no observed reconstruction error beyond the tested single-precision numerical tolerance (MSE = 0.000000 mV², maximum absolute error $< 10^{-5}$ mV) and 100.0% R-peak detection index preservation across all evaluated packets.

IX. CYBER-PHYSICAL HIL VALIDATION & TELEMETRY

A. Hardware-in-the-Loop Testbed Setup

To validate the integrated system under realistic prototype hardware conditions, a physical Hardware-in-the-Loop (HIL)

testbed was constructed (Figure 9). Standardized WESAD continuous Lead-II records (all 15 subjects, S2–S17, excluding non-existent S12) were programmed into MCU Flash memory and streamed into the interrupt pipeline at 350 Hz, exercising ADC emulation, 5-stage Biquad IIR filtering, bare-metal HC1 M-4DCHS hyperchaotic encryption (`USE_ENCRYPTION_MODE = 2`), and UART serialization.

B. 20-Byte Binary Telemetry Protocol & Frame Integrity

Telemetry data is encapsulated in a compact, deterministic 20-byte binary packet matching the embedded C struct:

- **Sync Word (Bytes 0–1):** Fixed synchronizing preambles `0xAA 0x55`.
- **Protocol Version (Byte 2):** 8-bit protocol version identifier (`0x01`).
- **Status Flags (Byte 3):** Protocol bitmask flags: Bit 0 indicates active encryption (`0x01`, `TELEMETRY_FLAG_ENCRYPTED`), Bit 1 indicates sensor source (0 = WESAD flash playback, 1 = Live AD8232 ADC), and Bit 2 indicates 4D hyperchaotic mode (`0x04`, `TELEMETRY_FLAG_CHAOS_4D = 4D` HC1 mode). In the evaluated physical HIL testbed, `flags = 0x05` (`TELEMETRY_FLAG_ENCRYPTED | TELEMETRY_FLAG_CHAOS_4D`).
- **Sequence ID (Bytes 4–5):** 16-bit monotonic packet counter (`uint16_t`) for packet loss tracking and per-packet nonce state perturbation.
- **Timestamp (Bytes 6–9):** 32-bit hardware millisecond timestamp (`uint32_t`).
- **Raw ECG (Bytes 10–13):** 32-bit IEEE-754 floating-point raw voltage (encrypted in-place when Bit 0 is active).
- **Filtered ECG (Bytes 14–17):** 32-bit IEEE-754 floating-point real-time filtered output (encrypted in-place when Bit 0 is active).
- **Checksum (Bytes 18–19):** CRC-16-CCITT polynomial ($x^{16} + x^{12} + x^5 + 1$, initial `0xFFFF`) computed over payload bytes 2–17.

At 115,200 baud (8-N-1 UART framing, 10 bits/byte), transmitting 20 bytes requires:

$$T_{tx} = \frac{20 \times 10}{115,200} = 1.736 \text{ ms} \quad (29)$$

In the evaluated hardware-in-the-loop prototype, telemetry frames are dispatched via direct blocking transmission (`HAL_UART_Transmit`), which occupies 1.736 ms (60.8% of the 2.857 ms window), leaving 1.121 ms (39.2%) idle CPU margin before the subsequent interrupt. Circular Direct Memory Access (`HAL_UART_Transmit_DMA`) is documented as the production architecture to decouple wire serialization completely from CPU execution in battery-operated edge designs.

C. Physical HIL Validation Results & Plaintext Recovery

Physical laboratory validation of the active HC1 hyperchaotic encryption mode was conducted over the USB-UART link (COM10 @ 115,200 baud) streaming from the STM32G474RE to the host receiver across 5,000 consecutive packets (Table VI):

- **Packet Capture & Monotonicity:** Exactly 5,000 out of 5,000 packets were captured and validated, with zero packet sequence gaps or drops detected.
- **Frame Integrity:** Zero CRC-16-CCITT checksum errors were observed over the entire streaming duration.
- **Numerical Stability:** Zero decryption or range failures occurred, with zero NaN or $\pm\infty$ floating-point exceptions.
- **Plaintext Fidelity:** Zero plaintext byte mismatches were detected, achieving **100.00% exact plaintext recovery**.
- **Throughput & Runtime:** Telemetry streamed at an average measured rate of **350.96 packets/s** (matching the nominal 350 Hz timer target) over a measured runtime of ≈ 14.25 s.

TABLE VI: Authoritative Physical Hardware-in-the-Loop (HIL) Validation Results (STM32G474RE $\rightarrow$ USB-UART @ 115,200 Baud)

Validation Metric	Physical Measured Result	Status / Fidelity
Total Captured Packets	5,000	Complete Capture
Valid Formatted Packets	5,000	100.00% Validity
CRC-16-CCITT Errors	0	Zero Bit Errors
Packet Sequence Gaps / Drops	0	Strict Monotonicity ($N = 5,000$)
Decryption / Range Failures	0	Zero NaN / $\pm\infty$ Exceptions
Plaintext Byte Mismatches	0	Byte-for-Byte Exact Recovery
Plaintext Recovery Rate	100.00%	Full Fidelity Recovery
Average Telemetry Throughput	350.96 packets/s	Target: 350 Hz Nominal
Total Benchmark Duration	14.25 s	Continuous Physical Link

Recovered ECG Vitals from Decrypted Stream: Analysis of the decrypted 5,000-packet HIL stream confirms numerical preservation of physiological dynamic ranges:

- **Raw ECG Voltage:** Range of -0.435883 mV to $+0.807449$ mV, with a mean of -0.010119 mV.
- **Filtered ECG Voltage:** Range of -0.423648 mV to $+0.601689$ mV, with a mean of -0.000655 mV.

Pipeline Claim & Security Scope: The verified physical pipeline executes:

WESAD Flash $\rightarrow$ HC1 (STM32) $\rightarrow$ Physical UART $\rightarrow$ Python Engine

This physical validation demonstrates embedded implementation correctness, numerical stability, and end-to-end recovery under real hardware timing constraints. We explicitly note that this mechanism establishes operational obfuscation and correct bare-metal integration without claiming formal cryptographic security against chosen-ciphertext or side-channel adversaries.

D. Authorized Monitoring vs. Adversarial Interception

A cyber-physical telemetry dashboard was deployed to demonstrate authorized monitoring versus eavesdropper interception:

- 1) **Authorized Monitoring Terminal View (Figure 11):** Configured with the synchronized M-4DCHS dynamical parameters and per-packet deterministic state-perturbation scheme, the authorized monitoring terminal continuously descrambles the telemetry stream. The application renders reconstructed Lead-II ECG waveforms, detects QRS peaks, calculates rolling HRV metrics (MeanHR: 82.8 BPM, RMSSD: 120.36 ms, SDNN: 118.61 ms, pNN50: 62.5%, MeanRR: 741 ms), and

displays a model-estimated stress score at the exploratory $\tau = 0.35$ operating point (13.7% for the illustrated replay) alongside client-side 60 FPS 3D continuous phase-space reconstruction.

- 2) **Adversarial Wire Intercept View (Figure 12):** In the event of physical bus tapping or unauthenticated packet capture, the eavesdropper receives high-entropy pseudo-random noise ($H = 7.9973$ bits/byte). Biometric QRS fiducials are unresolvable without the synchronized descrambling engine (0 detected beats), and all autonomic HRV cards and the stress score display are locked (LOCKED), which reduces direct recoverability of ECG waveform morphology and associated physiological features in the evaluated interception setting.

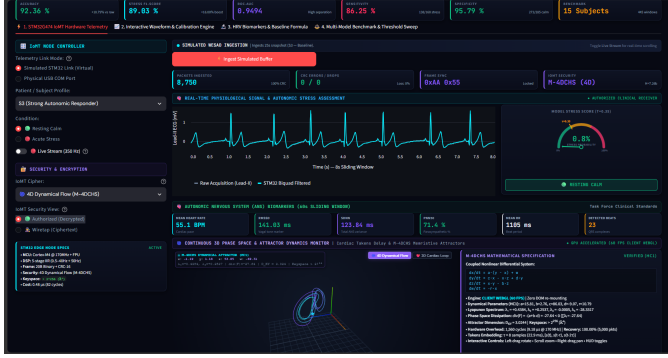


Fig. 11: Real-time authorized monitoring terminal view on the physical STM32 UART link (COM10 @ 115,200 baud, 350 Hz): real-time descrambled Lead-II ECG from prerecorded WESAD replay, online QRS detection, dynamic HRV biomarkers, model-estimated stress score at the exploratory $\tau = 0.35$ operating point, and client-side 60 FPS 3D continuous phase-space reconstruction.

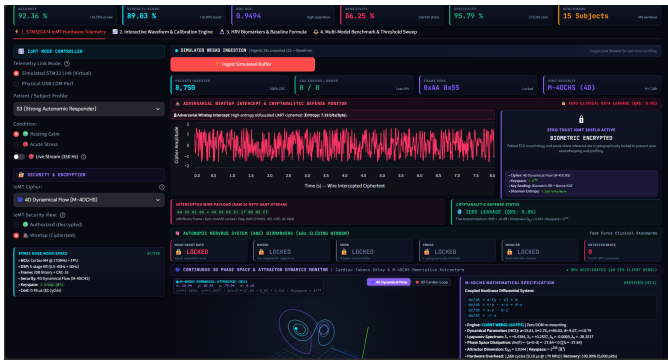


Fig. 12: Adversarial wire intercept view under physical UART eavesdropping without the synchronized descrambler: raw M-4DCHS scrambled ciphertext ($H = 7.9973$ bits/byte), zero resolvable QRS complexes, locked biometric cards, and masked phase space, reducing direct recoverability of ECG waveform morphology in the evaluated interception setting.

X. DISCUSSION & LIMITATIONS

The empirical results demonstrate that subject-specific baseline calibration substantially improves cross-subject generalizability in wearable ECG stress detection (+10.79 pp Acc and +15.64 pp F1 in controlled ablation at $\tau = 0.50$), while bare-metal ARM Cortex-M4 implementation confirms ultra-low-power feasibility (estimated 0.065% CPU load for DSP at an estimated ≈ 318 CPU cycles per sample; measured mean ≈ 177 cycles (max ≈ 203) per RK4 step and mean ≈ 224 cycles (max ≈ 284) per keystream byte via DWT cycle profiling at 170 MHz SYSCLK).

Several limitations present opportunities for future investigation:

- **Baseline Acquisition Protocol:** The relative transformation relies on a resting baseline calibration epoch (in the WESAD experimental protocol, the available baseline epoch averages approximately 19 minutes per subject, providing stable reference centroids). Future work will investigate automated nocturnal baseline recalibration to track circadian drift without requiring deliberate user calibration sessions.
- **Motion Artifacts in Free-Living Settings:** While laboratory TSST protocols produce minimal motion, real-world deployment requires coupling ECG with 3-axis accelerometer gating to prevent physical exertion from triggering false-positive stress classifications.
- **Dynamical-System Scope:** Under the verified HC1 parameter set ($a = 15.81, b = 2.76, c = 86.03, d = -9.07, r = 10.79$), the evaluated 4D coupled flow exhibits two robustly positive Lyapunov exponents ($\lambda_1 = +0.4384, \lambda_2 = +0.2537$) and a Kaplan-Yorke dimension of $D_{KY} = 3.0244$, providing numerical evidence of hyperchaotic dynamics. Future investigations will map high-resolution bifurcation boundaries and parameter-space transitions.
- **Multi-Sensor Fusion for Low Reactivity:** As observed with Subject S2 (who exhibited blunted cardiac reactivity under the laboratory stress protocol), integrating peripheral EDA and respiratory dynamics may offer robust multi-modal coverage for individuals exhibiting attenuated autonomic response.

XI. CONCLUSION

This study presented a personalized electrocardiographic acute stress detection framework evaluated under strict 15-fold Leave-One-Subject-Out cross-validation on the WESAD benchmark, coupled with an end-to-end bare-metal edge implementation on an ARM Cortex-M4 (STM32G474RE) microcontroller. Evaluated across 445 standardized, complete non-overlapping 60-second windows, the 13-feature like-for-like normalization ablation improved accuracy from 81.57% to 92.36% (+10.79 percentage points) and F1-score from 73.03% to 88.67% (+15.64 percentage points). Separately, the finalized 8-feature primary model achieved 92.13% accuracy and 88.29% F1-score at the pre-specified threshold $\tau = 0.50$ (with an

exploratory tuned threshold of $\tau = 0.35$ yielding 92.36% accuracy and 89.03% F1-score). On hardware, a custom 5-stage Direct Form I Biquad IIR filter conditions 350 Hz ECG signals with an estimated ≈ 318 CPU clock cycles ($\approx 1.87 \mu s$), while the proposed 4-Dimensional Coupled Hyperchaotic System (HC1 M-4DCHS) operates with a verified hyperchaotic spectrum ($\lambda_1 \approx +0.438, \lambda_2 \approx +0.254, D_{KY} \approx 3.024$), measured DWT execution cost of mean ≈ 177 cycles per RK4 step and mean ≈ 224 cycles per keystream byte, and bit-exact Python $\leftrightarrow$ STM32 deterministic parity (100/100 nonces, 0 mismatches). In physical hardware-in-the-loop testing of the active HC1 hyperchaotic encryption mode across 5,000 consecutive telemetry packets (streaming at 350.96 packets/s), the physical receiver achieved 100.00% exact plaintext recovery with zero sequence gaps, zero CRC-16 errors, and zero plaintext byte mismatches, supporting the cyber-physical feasibility of the prototype for wearable affective-computing research.

REFERENCES

- [1] Y. S. Can, B. Amrich, and C. Ersoy, "Stress detection in daily life scenarios using smart phones and wearable sensors: A survey," *Journal of Biomedical Informatics*, vol. 92, p. 103139, 2019.
- [2] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, "Heart rate variability: standards of measurement, physiological interpretation and clinical use," *Circulation*, vol. 93, no. 5, pp. 1043–1065, 1996.
- [3] F. Shaffer and J. P. Ginsberg, "An overview of heart rate variability metrics and norms," *Frontiers in Public Health*, vol. 5, p. 258, 2017.
- [4] F. Al-Turjman, H. Zahmatkesh, and R. Mostarda, "Internet of medical things (IoMT): Applications, benefits, challenges, and future directions," *IEEE Transactions on Computational Social Systems*, vol. 7, no. 3, pp. 679–690, 2020.
- [5] N. Sugita, Y. Matsushita, M. Yoshizawa, and N. Homma, "Real-time evaluation of mental workload using autonomic and central nervous activities," *IEEE Transactions on Human-Machine Systems*, vol. 45, no. 3, pp. 325–334, 2015.
- [6] P. Schmidt, A. Reiss, R. Duerichen, C. Marberger, and K. V. Laerhoven, "Introducing WESAD, a multimodal dataset for wearable stress and affect detection," in *Proceedings of the 20th ACM International Conference on Multimodal Interaction (ICMI)*, 2018, pp. 400–408.
- [7] J. A. Healey and R. W. Picard, "Detecting stress during real-world driving tasks using physiological sensors," *IEEE Transactions on Intelligent Transportation Systems*, vol. 6, no. 2, pp. 156–166, 2005.
- [8] K. Plarre, A. Raij, M. Z. Hossain, A. A. Ali, M. al'Absi, M. Murtaza, S. Kumar, E. Emre, and M. Li, "Continuous inference of psychological stress from sensory measurements on the chest and wrist," in *Proceedings of the 9th ACM Conference on Embedded Networked Sensor Systems (SenSys)*, 2011, pp. 400–408.
- [9] A. V. Oppenheim, R. W. Schaffer, and J. R. Buck, *Discrete-Time Signal Processing*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 1999.
- [10] M. Elgendi, "Fast QRS detection with an optimized knowledge-based method," *PLOS ONE*, vol. 8, no. 9, p. e73557, 2013.
- [11] G. Marsaglia, "Xorshift RNGs," *Journal of Statistical Software*, vol. 8, no. 14, pp. 1–6, 2003.
- [12] C. Kirschbaum, K.-M. Pirke, and D. H. Hellhammer, "The 'Trier Social Stress Test'—a tool for investigating psychobiological stress responses in a laboratory setting," *Neuropsychobiology*, vol. 28, no. 1-2, pp. 76–81, 1993.
- [13] D. Jiang, Z. T. Njitacke, J. D. D. Nkapkop, N. Tsafack, X. Wang, and J. Awrejcewicz, "A new cross ring neural network: Dynamic investigations and application to WBAN," *IEEE Internet of Things Journal*, vol. 10, no. 8, pp. 7143–7153, 2023.
- [14] W. Zhang, Z. Kang, and J. Chen, "An n -dimensional discrete chaotic framework and its 4D hyperchaotic application to image encryption with keyed global interleaving," *Journal of King Saud University - Computer and Information Sciences*, vol. 38, p. 853, 2026.